BHARATHI SEKAR

Software Engineer ~ Generative AI Developer

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SUMMARY

Software Engineer with experience developing and improving software systems, from resolving application issues to building full-stack applications and data-processing workflows. Hands-on experience building LLM-powered applications using multi-agent orchestration, RAG, tool integration, access controls, and evaluation. I enjoy understanding how a system works, identifying what needs to change, and building practical solutions that are reliable, secure, and useful to users.

SKILLS

Languages	Python, JavaScript, TypeScript, SQL, HTML, CSS
AI	LLMs, RAG, Multi-Agent Systems, Conversational AI, Prompt Engineering, NLP, AI Evaluation
AI Tools	OpenAI API, LangChain, LangGraph, Hugging Face, GitHub Copilot, Claude Code
Frameworks	React, FastAPI, Flask, Pydantic
Cloud & DevOps	Azure, Docker, Kubernetes, CI/CD, Git
Database	MySQL, MongoDB, PostgreSQL, Vector Databases
Analytics	Power BI, Pandas, NumPy, Exploratory Data Analysis, Data Visualisation, Statistical Modelling

EXPERIENCE

6/2025 – 12/2025 **Research Assistant** — **Crowd Computing Research Group, UBICOMP, University of Oulu, Finland**

- Designed, implemented, and deployed a **multi-agent LLM** solution using the **OpenAI API** to improve resistance to misinformation through **conversational AI**.
- Built a **rule-based orchestration framework** for multi-agent conversations, implementing turn-taking, urgency scoring, speaker nomination, and decision logic to coordinate LLM agents.
- Developed reusable **prompting** and **context-management** components using structured **JSON** to support consistent, context-aware responses across multiple AI agents.
- Developed a full-stack web platform using **Python, JavaScript,** and **SQL**, integrating backend services, APIs, database functionality, session management, and automated task assignment for **150+ users**. Deployed using **Docker** on **CSC Cloud** and **Azure**.
- Evaluated system performance, agent-role identification, and user interactions across three experimental cycles, using the results to refine the system and improve agent-role identification accuracy from **65.2%** to **80.6%**.

10/2025 – 12/2025 **Teaching Assistant** — **University of Oulu, Finland**

Guided **130+** students in **Human-Computer Interaction** exercise class, teaching mobile application design using Figma and providing individual feedback.

7/2023 – 8/2024 **Junior Software Engineer** — **Cognizant Technology Solutions, India**

- Maintained **10+ business-critical financial and payroll applications** as a technical point of contact, resolving production issues and coordinating changes to prevent business disruption. Received recognition for maintaining application stability and minimising business impact.
- Enhanced existing **Python microservices** and **REST APIs** by adding features, implementing business requirements, resolving issues, and improving user workflows.
- Built a financial file management application using **React, Python,** and **REST APIs**, bringing reports, transaction data, cash data, and policy documents into a single application for easier access.
- Performed **SQL-based data extraction, validation,** and **transformation** to analyse financial data, trace discrepancies across records, and identify issues affecting reporting accuracy.
- Automated recurring **data-processing workflows** using **Power Automate** and **Power Query**, reducing manual work by approximately **4 hours per week** and improving process consistency.
- Supported cloud migration and application integration across **Azure**, coordinating with upstream and downstream teams to manage dependencies and minimise disruption during changes.

7/2022 – 7/2023 **Software Engineer Trainee** — **Cognizant Technology Solutions, India**

- Investigated and resolved application issues across the team's supported applications, analysing logs and service data to identify root causes and restore functionality promptly.
- Built web applications using **Python, TypeScript,** and **React**, following **CI/CD** workflows with deployments in **Azure** and **AWS** environments.
- Created **Power BI dashboards** to help business teams analyse banking and insurance application data and identify trends.

2/2022 – 6/2022 **Software Engineer Intern** **Cognizant Technology Solutions, India**

- Designed and developed a file transfer system for two internal teams, automating data transfer using **React, TypeScript, Python, SQL, and RESTful APIs**, with **Azure Active Directory** integration for authentication.
- Performed frontend testing using **Jest** and backend testing using **Pytest**.

PROJECTS

Multi-agent Chatbot System for Inoculation Against Misinformation **Thesis/ GitHub**

Designed and implemented a multi-agent conversational system with role-based LLM prompting using OpenAI API. Built rule-based turn-taking logic and context-aware response generation. Conducted a web-based user experiment to study how AI-driven conversations can inoculate users against misinformation and how users perceived the different chatbot roles.
Python · OpenAI API · LLMs · Multi-Agent Systems · Flask · SQL · JavaScript · CSC Cloud · Azure · Docker

IT Operations Agent **GitHub**

Built an AI-powered IT operations agent using RAG and LangGraph for multi-step orchestration and controlled knowledge retrieval. Integrated tools for incident investigation, employee and permission lookup, and ticket management, with authentication and role-based authorization using Azure. Developed a FastAPI and PostgreSQL backend, a React frontend, and a structured evaluation suite covering tool selection, access controls, security, incident handling, and RAG retrieval accuracy.
Python · Azure · LangGraph · RAG · Pinecone · OpenAI API · FastAPI · PostgreSQL · React · Agent Orchestration · Evaluation

Large-scale Automated AI-powered Data Crawling and Categorization System **GitHub**

Built an AI-powered data pipeline for automated web crawling, content extraction, categorization, translation from Finnish to English, and semantic retrieval. Developed scalable Python workflows to collect, clean, structure, and process large volumes of e-commerce data, integrating embeddings and vector search for LLM-powered analysis.
Python · RAG · Ollama · Embeddings · Qdrant · Google Translate · CSC Cloud · Azure

Pandemic Response & Contact Tracing Distributed System **GitHub**

Developed an architecture consisting of 8 containerized microservices, API-based communication, event-driven processing, and database integrations. Deployed the system on CSC Cloud using Docker and Kubernetes, with Prometheus and Grafana for monitoring. Integrated OpenAI for automated alert generation and dynamic response handling.
Python · Docker · Kubernetes · REST APIs · gRPC · Kafka · PostgreSQL · MongoDB · OpenAI · Pytest

EDUCATION

9/2024 - 12/2025 **Master of Science (Technology), Computer Science and Engineering** **University of Oulu, Finland**
Major: Applied Computing, with a focus on software systems, data analytics, applied AI, and system analysis.

7/2018 - 5/2022 **Bachelor of Technology, ECE** **SRM Institute of Science and Technology, India**
Focused on signal processing, embedded systems, wireless communication, and software development.

CERTIFICATION

Microsoft Certified: Azure Fundamentals (AZ-900)

PUBLICATIONS

Co-author, **Conversational Inoculation to Enhance Resistance to Misinformation**, CHI 2026 Workshop on Envisioning the Future of Interactive Health, arXiv:2601.22394 - **Honourable Mention Award**

LANGUAGES

English - C2, **Finnish** - A2

REFERENCES

Prof. Simo Hosio, Associate Professor
University of Oulu
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Research Supervisor (during Research Assistant position)